A2

ns-3 Setup & 3GPP XR Pose Traffic Model

A deterministic uplink baseline based on 3GPP TR 26.926 §5.9.2

[Your Name] · [Student ID]

Week 7 · 10-minute presentation

ns-3.44 | Ubuntu 22.04.5 LTS | x86-64

ENVIRONMENT

Setup and reproducibility

ns-3.44
Simulator version

Ubuntu 22.04.5
Linux platform · x86-64

60 s
Traffic duration · 6,000 packets

Build and run

```
$ ./ns3 build xr-pose-uplink
$ ./ns3 run xr-pose-uplink
$ python3 scratch/plot-xr-pose-results.py --prefix xr-pose-uplink
```

What broke?

Restricted ccache directory
→ built with CCACHE_DISABLE=1

Matplotlib GUI warnings
→ switched plotting backend to Agg

3GPP Pose uplink model

Specified by §5.9.2

- Pose is available to the application every 1 ms (1 kHz).
- Each IP packet is 100 bytes: 60-byte payload + 40-byte headers.
- The latest pose is reported every 10 ms (100 Hz).
- Target uplink latency is no greater than 10 ms.

Experimental choices

- UDP/IPv4 over a point-to-point baseline link.
- One XR device and one application server.
- Fixed pose: position (1,1,1.2), quaternion (0,0,0,1).
- The timestamp changes every sample; motion is not required.

Source: 3GPP TR 26.926 V18.2.0, §5.9.2 — Pose Uplink Model

Topology and 100-byte packet format

XR DEVICE
10.1.1.1

UDP/IPv4 →
10 Mbps · 1 ms one-way

SERVER
10.1.1.2

100-byte IPv4 packet



Pose payload: position 12 B · quaternion 16 B · sample time 8 B · device ID 4 B · reserved 20 B

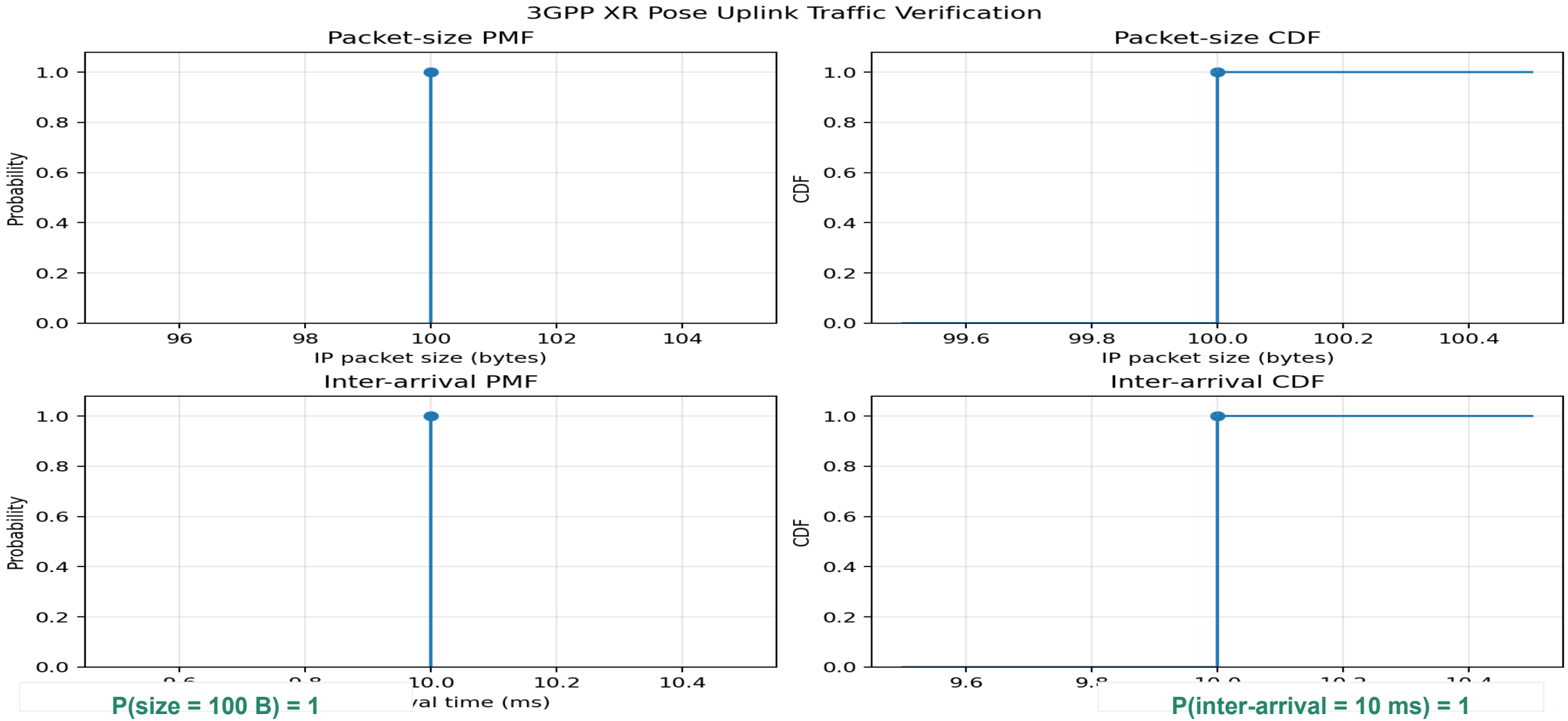
Sequence 4 B + send time 8 B

Network delay = receive time − send time

Pose age = receive time − sample time

PCAP adds a 2-byte PPP header: captured frame = 102 B

Traffic-model verification: PMF and CDF

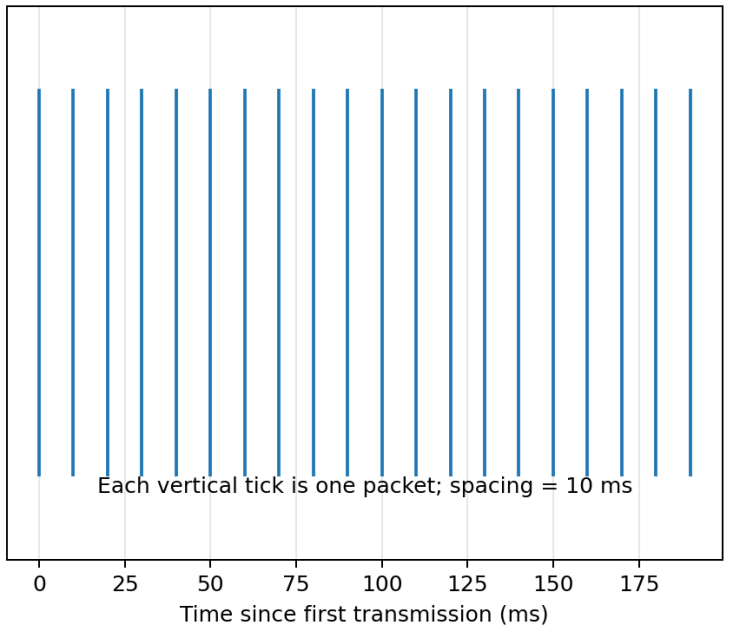


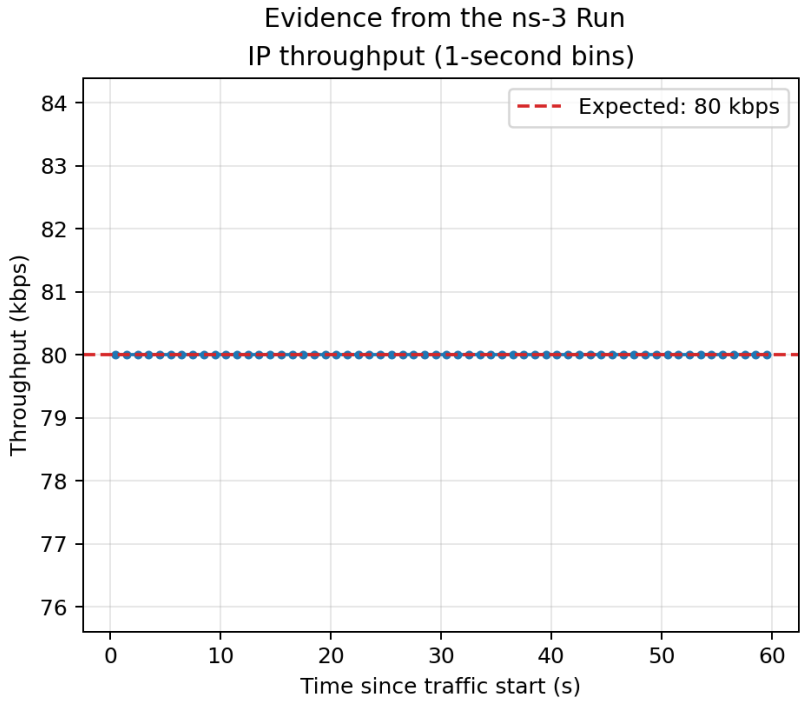
P(size = 100 B) = 1

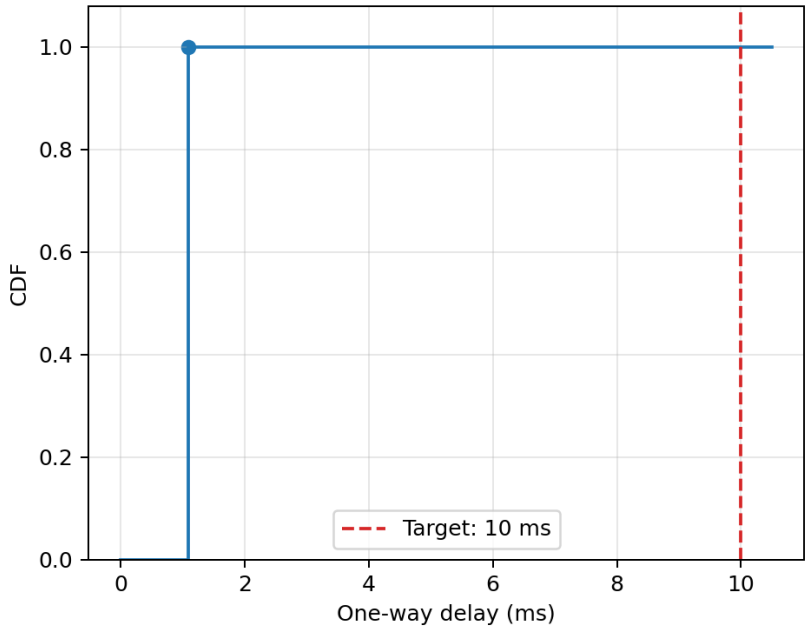
P(inter-arrival = 10 ms) = 1

RESULTS

Evidence from my ns-3 run







80 kbps
Measured IP offered rate

1.0816 ms
Mean and maximum delay

100%
Packets meeting ≤10 ms

PCAP proof and packet accounting

```
1.000000   IP (0x0021), length 102: 10.1.1.1 → 10.1.1.2   UDP, length 72
1.010000   IP (0x0021), length 102: 10.1.1.1 → 10.1.1.2   UDP, length 72
1.020000   IP (0x0021), length 102: 10.1.1.1 → 10.1.1.2   UDP, length 72
```

What to verify in Wireshark

- Filter: udp.port == 5000
- Time delta between packets: 0.010 s
- IPv4 total length: 100 bytes
- UDP length: 80 bytes; UDP payload: 72 bytes

Why does PCAP show 102 B?

2 B + **100 B**

PPP header — IPv4 packet

= 102-byte captured frame

TAKEAWAYS

Conclusion

The implemented traffic model matches the configured 3GPP baseline.

6,000 / 6,000
Packets sent / received · 0% loss

100 B · 10 ms
Deterministic size and reporting period

1.0816 ms
Well below the 10 ms latency target

Why UDP?

Pose updates become stale quickly. UDP avoids retransmission, acknowledgements, and head-of-line blocking; the next fresh pose is more useful than a delayed old pose.

Next step

Add movement, wireless loss, jitter, or congestion only after the baseline traffic model is verified.

Questions?